Quantization of gauge system

1. Introduction of constrained system.

Origin of constraints: (1) "primary constraint" comes from Lagrangian and the definition of canonical momentum $\Pi_\ell(x) \equiv \frac{\partial \mathcal{L}[\Phi(x), \partial_\mu \Phi(x)]}{\partial(\partial_0 \Phi^\ell(x))}$

eg. EM field $\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + A^\mu J_\mu$

$\Pi^\mu(x) = F^{\mu 0}(x)$, thus $\Pi^0(x) = 0$ which is canonical conjugate of $A^0(x)$

Why it is a problem?

$\Rightarrow$ Generally, we will have ill-defined commutator $[A^0(x), \Pi^0(y)]_{x^0=y^0} = i\delta^{(3)}(\mathbf{x}-\mathbf{y})$

we say $\Pi^0(x) \approx 0$ is primary constraint.

(2) "secondary constraint" comes from the requirement of primary constraint being consistent with e.o.m.

eg. one observes that there is no time evolution term of $A^0$ in our Lagrangian which is a consequence of primary constraint $\Pi^0(x) \approx 0$

Thus $\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_0)} - \frac{\partial \mathcal{L}}{\partial A_0} = 0 \Rightarrow \partial_\mu (F^{0\mu}) - J^0 = 0.$

The distinction of "primary" and "secondary" is of no importance in the following discussion because it is merely depended on our choice of Lagrangian.

2. First class constraint and gauge transformation.

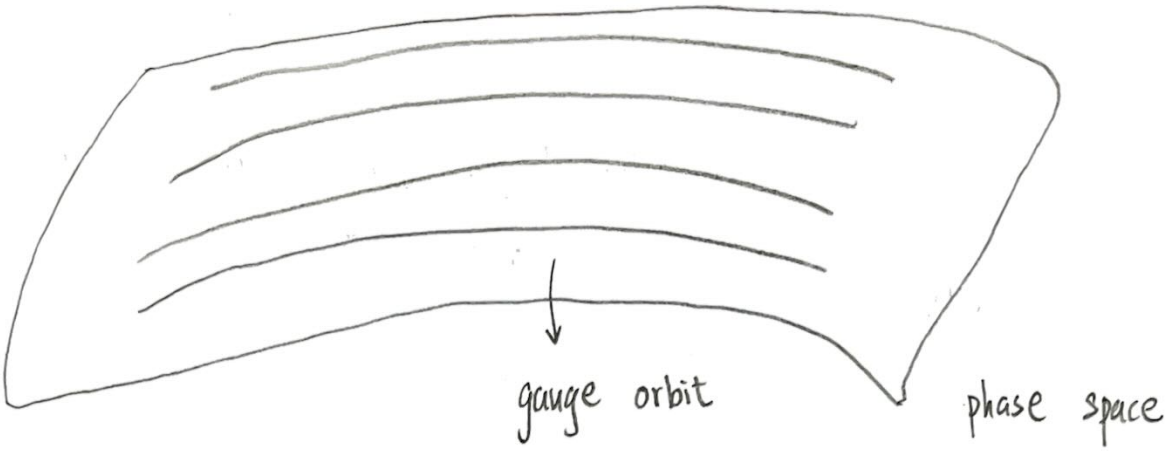
Instead we classify constraints by Poisson bracket $[X, Y]_P \equiv \frac{\partial X}{\partial \Phi_{\ell x}} \frac{\partial Y}{\partial \Pi^\ell_x} - \frac{\partial Y}{\partial \Pi^\ell_x} \frac{\partial X}{\partial \Phi_{\ell x}}$

if Poisson brackets of $\hat{a}$ given constraint with all the other constraints vanish $[\chi_M, \chi_N]_P = 0$ for all $N$

$\chi_M$ is called first class constraint.

One important property of first-class constraints is that it is the generator of trivial transformation (gauge transformation) that doesn't affect the physical state.

c.f. Dirac Lectures on Quantum Mechanics, here we provide geometric meaning




gauge orbit: A collection of points in phase space which a given point in configuration space mapped to via Legendre transformation.

$$H(q,p) = p\dot{q} - L(q,\dot{q})$$

$\Rightarrow$ example we can see this via discussing dependence of $p$ on $q$.

$$p_i \equiv \frac{\partial L}{\partial \dot{q}_i} \qquad \delta p_i = \frac{\partial^2 L}{\partial q_j \partial \dot{q}_i}\delta q_j + \underbrace{\boxed{\frac{\partial^2 L}{\partial \dot{q}_j \partial \dot{q}_i}}}_{\equiv M_{ij}} \delta \dot{q}_j$$

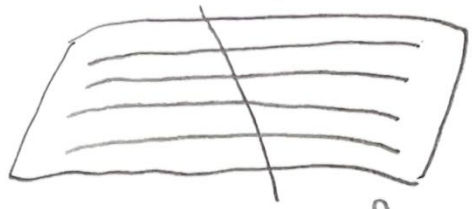
if $M_{ij}$ is not invertible $\Rightarrow$ there exists non-empty nullspace

$h_i M_{ij} = 0$, for some $\vec{h} \neq \vec{0}$

Then we got $h_i \delta p_i = h_i \frac{\partial^2 L}{\partial q_j \partial \dot{q}_i}\delta q_j$, showing $p$ depending only on $q$ at projection to the null space!

$q$ and $p$ aren't independent variables thus Hamiltonian equation of motion is ill-defined!

We are saved by choosing a gauge (gauge fixation)



gauge fixation.

Which should intersects with gauge orbits once and only once

There are constraints whose Poisson bracket with other constraints doesn't vanish. $[X_M, X_N]_P \neq 0$ for some N.

which called second class constraint.

When all first class constraints being eliminated by choosing a gauge.

the matrix of Poisson brackets of the remaining constraints. is non-singular.

$$\det C \neq 0 \qquad C_{MN} \equiv [X_M, X_N]_P.$$

Dirac suggests the commutator now is

$$[X, Y] = i\,[X, Y]_D \rightarrow \text{so-called Dirac bracket.}$$

$$[X, Y]_D = [X, Y]_P - \sum_{M,N} [X, \phi_M]_P C^{-1}_{MN} [\phi_N, Y]_P \qquad C^{-1}_{MN} \times C_{NP} = \delta_{MP}$$

where M, N go over second-class constraint.

☆ Quantization of EM Field.

$\Pi^0(x) \approx 0.$ compute Poisson bracket. $[\partial_i F^{0i}(x) - J^0(x), \Pi^0(y)]_P$

$\partial_i F^{0i} - J^0 \approx 0.$ $= [-\partial_i \Pi^i(x) - J^0(x), \Pi^0(y)]_P = 0.$

they are first-class constraints.

which means we cannot uniquely determine constrained variable $A^0(x)$

and therefore leave others $A^i(x)$ as canonical variables.

Thm. the Dirac bracket is equal to the Poisson bracket calculated in terms of the reduced set of unconstrained canonical variables $Q^n, P^n$.

by Maskawa and Nakajima.

Let's choose Columb gauge. $\nabla\cdot A(x)=0$.

plug this gauge fixation into $\partial_i F^{0i} - J^0 = 0$.

$$\partial_i\left(\partial^0 A^i(x) - \partial^i A^0(x)\right) - J^0(x) = 0.$$

$$\partial_i\partial^i A^0(x) + J^0(x) = 0.$$

$$\partial_i\partial^i \int \frac{d^3k}{(2\pi)^3}\tilde{A}^0(k)e^{-i\vec{k}\cdot\vec{x}} + \int\frac{d^3k}{(2\pi)^3}\tilde{J}^0(k)e^{-i\vec{k}\cdot\vec{x}} = 0.$$

$$-\int\frac{d^3k}{(2\pi)^3}\tilde{A}^0(k)\,k^2\,e^{-i\vec{k}\cdot\vec{x}} + \int\frac{d^3k}{(2\pi)^3}\tilde{J}^0(k)e^{-i\vec{k}\cdot\vec{x}} = 0. \qquad \tilde{J}^0(k) = \int d^3y\, J^0(y)e^{i\vec{k}\cdot\vec{y}}$$

$$A^0(x) = \int\frac{d^3k}{(2\pi)^3}\frac{\tilde{J}^0(k)}{k^2}e^{-i\vec{k}\cdot\vec{x}} = \int\frac{d^3k}{(2\pi)^3}\frac{1}{k^2}e^{-i\vec{k}\cdot\vec{x}}\int d^3y\, J^0(y)e^{i\vec{k}\cdot\vec{y}}$$

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which determines $A^0(x)$ uniquely

Now we can compute again $[\nabla\cdot A(x), \partial_i F^{0i}(y) - J^0(y)]_P$

$$= [\partial_i A^i(x), -\partial_i\Pi^i(y) - J^0(y)]_P$$

$$= +\partial_i^2\delta^{(3)}(x-y)$$

$$-\frac{\delta\partial_i A^i(x)}{\delta A^i(z)}\cdot\frac{\delta\partial_i\Pi^i(y)}{\delta\Pi^i(z)} = -\partial_i\left(\frac{\delta A^i(x)}{\delta A^i(z)}\cdot\frac{\delta\partial_i\Pi^i(y)}{\delta\Pi^i(z)}\right) + \frac{\delta A^i(x)}{\delta A^i(z)}\partial_i\frac{\delta\partial_i\Pi^i(y)}{\delta\Pi^i(z)}$$

$$= -\partial_{ix}\left(\delta^{(3)}(x-z)\,\partial_{iy}\delta^{(3)}(y-z)\right)$$

$$= -\partial_{ix}\delta^{(3)}(x-z)\,\partial_{iy}\delta^{(3)}(y-z)$$

$$= -\partial_{ix}\delta^{(3)}(x-z)\cdot(-\partial_{ix}\delta^{(3)}(y-x)) = +\partial_i\partial^i\delta^{(3)}(x-y)$$

let $\phi_{1x} = \partial_i A^i(x)$

$\phi_{2y} = \partial_i\Pi^i(y) + J^0(y)$

Thus the canonical variable $A^i(x,t), \Pi^i(y,t)$

$$[A^i(x), \partial_i A^i(y)]_P = 0 \qquad [A^i(x), \partial_i\Pi^i(y) + J^0(y)]_P = \frac{\delta A^i(x)}{\delta A^i(z)}\frac{\delta\partial_i\Pi^i(y)}{\delta\Pi^i(z)}$$

$$= \delta^{(3)}(x-z)\,\partial_{iy}\delta^{(3)}(y-z)$$

$$[\Pi^i(x), \partial_i\Pi^i(y) + J^0(y)]_P = 0$$

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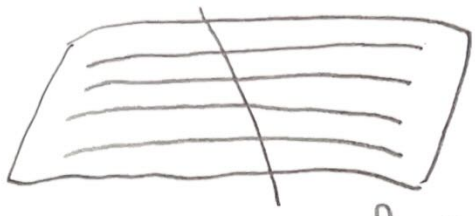

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Commutators of canonical variables $A^i(x)$, $\Pi^i(x)$

$$(C^{-1})_{1\mathbf{x},2\mathbf{y}} = [\Phi_{1\mathbf{x}}, \Phi_{2\mathbf{y}}]_P = -\int \frac{d^3p}{(2\pi)^3} \frac{1}{\mathbf{p}^2} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = -\frac{1}{4\pi|\mathbf{x}-\mathbf{y}|}$$

$$(C^{-1})_{2\mathbf{y},1\mathbf{x}} = [\Phi_{2\mathbf{y}}, \Phi_{1\mathbf{x}}]_P = -(C^{-1})_{1\mathbf{x},2\mathbf{y}}$$

$$\partial_i^2 \delta^{(3)}(\mathbf{x}-\mathbf{y}) = \partial_i^2 \int \frac{d^3k}{(2\pi)^3} e^{-i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}$$

$$= \int \frac{d^3k}{(2\pi)^3} (-\mathbf{k}^2) e^{-i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}$$

$$(C^{-1})_{1\mathbf{x},1\mathbf{y}} = 0 \quad (C^{-1})_{2\mathbf{x},2\mathbf{y}} = 0.$$

Thus $[A^i(\mathbf{x}), \Pi_j(\mathbf{y})]_P = i($

$$[A^i(\mathbf{x}), A_j(\mathbf{y})] =$$

$$[\Pi^i(\mathbf{x}), \Pi_j(\mathbf{y})] =$$

Mode expansions of $A^\mu(x)$

$$A^\mu(x) = \int \frac{d^3p}{(2\pi)^3\sqrt{2p^0}} \sum_\sigma \left( \epsilon^\mu(\mathbf{p},\sigma)\, a(\mathbf{p},\sigma)\, e^{-ip\cdot x} + \epsilon^{*\mu}(\mathbf{p},\sigma)\, a^\dagger(\mathbf{p},\sigma)\, e^{ip\cdot x} \right)$$

$A^0(x)$ isnot independent field variable but made off by others canonical fields.

$\Rightarrow$ $\varepsilon^0(\mathbf{p},\sigma) = 0.$

$p_i \varepsilon^i(\mathbf{p},\sigma) = 0.$ from $\partial_i A^i(x) = 0.$

$\Rightarrow$ photon is spin 1 massless particle with 2 DOF, $\sigma = \pm 1$

and we can also prove from commutator relation of $A^\mu$ and $\Pi_\mu$.

$$\Rightarrow \sum_\sigma \epsilon^i(\mathbf{p},\sigma)\, \epsilon_j(\mathbf{p},\sigma) = \delta^i_j - \frac{p^i p_j}{|\mathbf{p}|^2}$$

$a(\mathbf{p},\sigma)$ and $a^\dagger(\mathbf{p},\sigma)$

$$[a(\mathbf{p},\sigma), a^\dagger(\mathbf{q},\sigma')] = \delta_{\sigma\sigma'}\, \delta^{(3)}(\mathbf{p}-\mathbf{q})$$

first-class constraints $\Pi^0(x) = 0$ $\quad \phi_1$

$$-\partial_i \Pi^i(x) - J^0(x) = 0 \quad \phi_2$$

for $\Pi^0(x) = 0$

$$\delta_1 A^\mu(\mathbf{x}, t) = \int d^3y\, \epsilon_1(\mathbf{y}, t) \left[ A^\mu(\mathbf{x}, t), \Pi^0(\mathbf{y}, t) \right]_P$$

$$= \int d^3y\, \epsilon_1(\mathbf{y}, t)\, \delta^\mu{}_0\, \delta^{(3)}(\mathbf{x} - \mathbf{y})$$

$$= \delta^\mu{}_0\, \epsilon_1(\mathbf{x}, t)$$

for $-\partial_i \Pi^i(x) - J^0(x) = 0$

$$\delta_2 A^\mu(\mathbf{x}, t) = \int d^3y\, \epsilon_2(\mathbf{y}, t) \left[ A^\mu(\mathbf{x}, t), -\partial_i \Pi^i(\mathbf{y}, t) - J^0(\mathbf{y}, t) \right]_P$$

$$= -\int d^3z \int d^3y\, \epsilon_2(\mathbf{y}, t) \frac{\delta A^\mu(\mathbf{x}, t)}{\delta A^i(\mathbf{z}, t)} \frac{\delta \partial_i \Pi^i(\mathbf{y}, t)}{\delta \Pi^i(\mathbf{z}, t)}$$

$$= \int d^3z \int d^3y\, \delta^{(3)}(\mathbf{y} - \mathbf{z})\, \partial_{iy} \left( \epsilon_2(\mathbf{y}, t) \delta^{(3)}(\mathbf{x} - \mathbf{z}) \right) \delta^\mu_i$$

$$= \delta^\mu_i\, \partial_i \epsilon_2(\mathbf{x}, t)$$

$$\star \int d^3z\, \delta^{(3)}(\mathbf{x} - \mathbf{y})\, \partial_{iy} \epsilon_2(\mathbf{y}, t)\, \delta^{(3)}(\mathbf{x} - \mathbf{z}) = \delta^{(3)}(\mathbf{x} - \mathbf{y})\, \partial_{iy} \epsilon_2(\mathbf{y}, t)$$

We see that $\epsilon\phi_1$ only generates change on $A^0(x)$

we can choose a subgroup such that $\epsilon_1(\mathbf{x}, t) = \partial_0 \epsilon(\mathbf{x}, t)$

and $\partial_i \epsilon(\mathbf{x}, t) = 0$. which matches $A^\mu(x) \to A^\mu(x) + \partial^\mu \epsilon(x)$

as for $\varepsilon\phi_2$, we choose $\partial_i \varepsilon_2(x,t) = \partial_i \varepsilon(x,t)$

$$\partial_0 \varepsilon(x,t) = 0.$$

which matches $A^\mu(x) \to A^\mu(x) + \partial^\mu \varepsilon(x)$ too.

Formal Procedure for Constrained System.

We know the problem arise from Lagrangian formalism and Hamiltonian formalism.

that $M^{ij} \equiv \frac{\partial^2 \mathcal{L}}{\partial \dot{q}_i \partial \dot{q}_j}$ is not full-rank

for N system. this M is $N \times N$ matrix, if M is $R<N$ rank.

$\Rightarrow$ there is $N-R$ velocities cannot be determined.

$$\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} = 0 \Rightarrow \frac{\partial \mathcal{L}}{\partial q_i} - \left( \frac{\partial^2 \mathcal{L}}{\partial q_j \partial \dot{q}_i} \dot{q}_j + \frac{\partial^2 \mathcal{L}}{\partial \dot{q}_j \partial \dot{q}_i} \ddot{q}_j \right) = 0.$$

And this implies there are $N-R$ constraints.

Example: $\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu}$

$$\frac{\partial^2 \mathcal{L}}{\partial \dot{A}_i \partial \dot{A}_j} \neq 0 \quad \text{but} \quad \frac{\partial^2 \mathcal{L}}{\partial \dot{A}_0 \partial \dot{A}_j} \neq 0 \Rightarrow M = \begin{pmatrix} 0 & 0 & \cdots & \cdots \\ 0 & - & \cdots & \cdots \\ 0 & \cdots & - & \cdots \\ \cdots & \cdots & \cdots & - \end{pmatrix}$$

$\Rightarrow$ 1 primary constraint.

Dirac field theory $\mathcal{L} = \bar{\psi}(i\partial\!\!\!/ - m)\psi$

Although few textbook mention it, but we can apply Dirac algorithm to Dirac field theory. We first assume $\psi^\dagger$ and $\psi$ are independent variables (just as we did in complex field theory)

$\Rightarrow$ it is obvions $M_{ij} \equiv \frac{\partial^2 \mathcal{L}}{\partial \psi_i^\dagger \partial \psi_j} = 0$. out of 2 rank.

$\Rightarrow$ this means there would be 2 primary constraints.

The rest analysis (Dirac bracket) is left for you!